

Supervision & Reflection Template

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What?

Reflect on what has happened in your project this month?

This month, I focused mainly on improving the backend and making the system more secure and stable rather than adding new features. One of the biggest things I did was finally moving the database from SQLite to PostgreSQL using Docker. This took a while to get working properly, but it's now set up and running as intended.

I also spent a lot of time improving security across the project. I added encryption for sensitive data, made sure things like JWT keys are handled properly through environment variables, and improved how authentication is handled. I also added rate limiting to protect routes like login and the vault feature, and made sure the database is not exposed unnecessarily.

Alongside this, I worked on testing and integration. I improved how errors are handled across the backend so responses are more consistent, added validation to make sure bad data can't be sent into the system, and wrote integration tests for key parts like authentication, rooms, appliances, and insights. I also fixed a number of issues that came up during this process, especially after switching to PostgreSQL.

So What?

This month was important because it made the project much more solid overall. Moving to PostgreSQL makes the system more realistic and closer to how a real application would be set up, especially compared to using a local file database. It also matches what I originally planned for the project .

The security improvements were also a big step forward. Instead of just having basic protection, the system now handles data more carefully and is better protected against common issues. This is important because security is a big part of the project's aim .

One of the main challenges this month was that a lot of things broke during the database migration. I had to fix configuration issues, bugs in how data was being loaded, and some backend errors before everything worked again. It wasn't the most visible progress compared to building features, but it was necessary to make sure the system is reliable going forward.

Now What?

Next, I need to move into Sprint 10 and Sprint 11 while also starting my technical documentation. Since most of the main features are now finished, the focus will be on testing, debugging, and polishing everything.

I'll continue improving test coverage, checking edge cases, and making sure all parts of the system work properly together. At the same time, I'll begin writing up the technical report, including how the system is designed, how security is handled, and how everything was tested. Breaking this into smaller sections should help me stay on track and avoid leaving too much until the end.

Student Signature

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